

AGC Executive

0.1 The restart table

The restart table is located in unswitched fixed storage and is organized into six restart groups. Within each group, one or more restart “phases” define the specific restart procedure within the group, each assigned to a unique restart location within the program. As a program progresses, and its restart requirements change, it informs the Executive through the PHASCHNG routine that a different restart phase is now in effect. As several different programs share the same group, an individual job might only use one, or perhaps a few, restart phases within the group. Each entry in the table defines a single recovery process, which is associated with only one program. There is no universal or all-encompassing restart table, and the contents in the Lunar Module version of the AGC software are considerably different from its Command Module counterpart. Four different restart options are available, and are defined through the attributes of the restart phase number. For clarity, it is easier to use the notation “G.P” to describe the restart point, with G as the group number, and P , the restart phase. Using this notation, the list of restart options are:

G. 0	Inhibit group restart
G. 1	Restart last display
G.Even#	Execute two restart routines
G.Odd# (> 1)	Execute one restart routine

In this notation, “even” phase numbers and “odd” phase numbers are exactly that; G.Even# include phases such as 2.2, 2.8, 4.6, and so forth. As G. 1 is already assigned, the “odd” phases must begin with 3 . An important effect of this convention is that both restart phases G. 0 and G. 1 are common to all groups, and neither requires defining a unique process to run. During a restart, each entry in the phase table is checked for a zero entry. If so, that restart group is not active and no additional processing occurs. Restart phase one redisplay the data on the DSKY, but does no other recovery processing. Only the remaining two types of restart phases, G.Even# and G.Odd# (> 1) will attempt to restart the program itself. Phases using odd numbers greater than one perform only one operation during that group’s restart processing, such as scheduling either a job or waitlist task. Two restart operations occur in even-number restart phases, and any combination of jobs or waitlist tasks are possible.

Each restart operation requires three words to define, with its contents depending on whether the restart action requires scheduling a job or a waitlist task. Oddnumber restart phases performing only one restart operation will only occupy three words, while even-numbered phases require six words to reflect the two operations scheduled for that group. Entries for a job in the restart table require three words: one word for the execution priority, and two words to define the entry point address. Waitlist tasks have a word dedicated to their scheduled start time, and also use two words for their entry address. Although this format appears sufficient, it does not contain all of the information required to schedule its work. In the quest to minimize the

amount of storage occupied by the phase table, the several flags necessary for specifying other program attributes are not stored in dedicated memory areas.

Rather, these flags are implemented as encodings of the three restart table words passed to PHASCHNG. In the case of a job, the first decision is to know if a VAC area is necessary or if only a Core Set will suffice. Using the priority word in the restart table entry, a positive priority value indicates that a VAC area is necessary. If the priority is complemented, a Core Set is obtained for the job but no VAC area. The entry point address is used in a similar manner. When complemented, it indicates that a waitlist task must be scheduled (not a job) and that the word used for defining the job's priority should be used as the waitlist scheduling time. This encoding scheme, which may appear on the surface to be needlessly complex, actually saves several words in the unswitched fixed storage area.

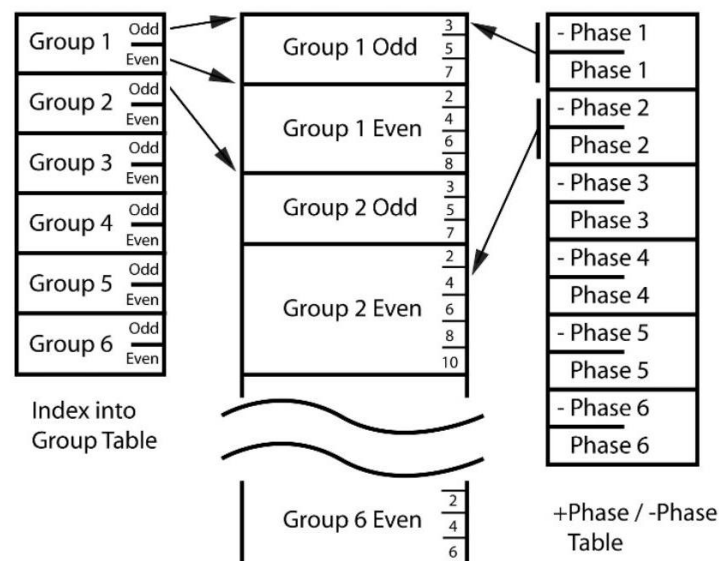


Figure 0.1: Restart and phase tables

0.2 Phase tables and restart processing

The number of options available with restart processing creates a dizzying array of data formats and parameter lists to process. However, the saving of storage space in the AGC is always a high priority, and the concise format of the PHASCHNG parameter lists reflects this. Each program protected by restart code has its own unique set of requirements, all of which must be passed to the PHASCHNG routine. At a minimum, the call to PHASCHNG must include the group and phase number, and flags to indicate important properties of the programs used for the restarts.

Two broad classes of parameters are used when calling PHASCHNG. Calls to PHASCHNG of "Type A" define a fixed, single-word parameter that references an explicit restart routine. "Type C" parameters include additional data words to describe variable data used by the restart routine. Such variable data might be the scheduling time for the defined recovery waitlist task, or the entry point address of the restart code. PHASCHNG also allows a third, "Type B", format, which is a hybrid that combines one fixed and one variable parameter. Highly condensed, the format of the data word following the call to PHASCHNG contains essential data such as the parameter type and group number. For those parameter lists that support variable data, bits also define what additional parameters are present.

0.3 The design of the phase tables

Dynamically altering the contents of the restart table is not possible, as it resides in fixed storage. This is reasonable, as the data in the restart table must be protected against any occurrence of a program failure that alters memory. However, restart routines frequently require data that is specific to the recovery needed, and this is saved in the phase table. The restart phase table contains the parameters necessary for recovering each restart group. The parameter strings passed to PHASCHNG are saved in the table, together with their bitwise complements. This odd means of storing data is an essential part of the sanity checking of the system during the recovery sequence. Saving the inverse of the data parameter is an effective means to assure that the restart tables have not suffered from corruption by a program error. Phase tables, like so many other critical data areas, are particularly vulnerable because they reside in unswitched (common) erasable storage. Without any storage protection mechanism in the AGC, damage to the phase tables can occur at any time and without any warning. Storing a second copy of the phase table might be useful,

but there is always the chance of having several words overlaid by identical garbage. For example, if an errant process wrote zeros throughout memory, how would this be detected without extensive logic? In a more general case, how do we determine if two identical copies of data are invalid? While the technique of saving both the data and its complement is far from fool-proof, it is trivial to implement and requires only a small amount of additional storage. Upon entering the restart processing, the phase table is checked to see if it is still valid. If a test of the phase table fails, erasable storage is presumed to be corrupted and program recovery stops. With a phase table that is longer valid, it is no longer possible to restart the program and so a hardware “fresh start” is initiated.

A check of the phase table’s health is driven by more than a conservative programming philosophy. When performing a restart, no effort is devoted to attempting to diagnose the root cause of the problem; rather, the restart process begins immediately. While this allows for a fast recovery, so important in a real-time system, it leaves open the question of how extensive the problem is. The most conservative assumption is that any routine could be at fault, so it is necessary to cancel all active jobs and tasks to ensure that the failing component is terminated. Logic errors are always likely suspects, and might generate a restart for any number of reasons. Of course, bad logic usually begets bad data, and it is highly likely that corrupted erasable memory resulted from the problem. Regularly examining several areas of erasable memory is part of the self-checking and restart processing. Unlike the phase table entries, testing of these data areas centers on checking for known values or ranges. If a discrepancy exists, it must be presumed that erasable storage is corrupted, and a fresh start is the only option to clear the error.

0.4 Phase tables part 2: control of Major Modes

A secondary function of phase table management routines is maintaining the program number of the currently executing Major Mode. This quantity, saved in the MODREG erasable storage location, is displayed on the DSKY as the program number. Maintaining the Major Mode number is little more than a housekeeping task, as a change in MODREG does not result in scheduling a new program. On the other hand, a number of routines check this value, and alter their logic based on the program that is currently running. In a few critical mission phases, the Major Mode progresses to a point where it starts to run significantly different execution logic. This change is presented to the crew as a change in the Major Mode number, making it appear as if a new program has started. For example, the progression of Major Modes during a lunar land-

ing (Program 63, then Program 64, and finally Program 65) is entirely automatic, saving the crew from the distraction of manually entering DSKY commands during a time of high workload. While the follow-on programs might appear as standalone Major Modes, their execution is dependent on the successful completion of the earlier stage.

0.5 A bad day at work: program alarms and restart processing

The AGC uses three different mechanisms to handle problems with the software or its inputs, recognizing the fact that not all situations require the same level of recovery. Program alarms, aborts and fresh starts are all possible options in the case of hardware or software problems, and each provides a different level of recoverability. Minor issues such as procedural errors may present themselves as program alarms, but do not require restart processing. In these cases, the program alarm is little more than an error message, such as an indication that the landing radar is not in the correct position, or that a program wishes to use the IMU while that hardware happens to be turned off. Other program alarms or aborts are a more serious matter. For a system resource shortage or a non-recoverable software error, the only recovery option is a restart of the AGC. Restarts from program aborts take on two different identities. A software restart terminates all running work and resets many of the system variables before performing recovery processing using the restart table. The objective of this type of restart is to preserve the execution environment and continue processing with a minimum of disturbance. It is important to realize that a software restart is not the same as “rebooting” a computer. Although work is thrown away, some restarts automatically reschedule programs for execution. More serious problems can result in cases where most of erasable memory is preserved but processing is halted in a so-called “P00DOO” abort. A far more drastic response to errors is the hardware restart, or fresh start. A fresh start clears all the work in the system, reinitializes the hardware and system variables, and places the system in an idle state. A fresh start is the equivalent of “rebooting” the AGC, with all work being lost and erasable memory reinitialized.

Each type of restart has a four-octal-digit code number associated with it that uniquely identifies the problem. A prefix digit to the code describes the type of recovery that is done: a 2 indicates that a P00DOO abort is called, and a 3 in the first digit signifies a software restart. Using this coding scheme, a 1202 alarm will be displayed as 31202 on the DSKY. Invoking a program alarm requires calling the **ALARM** executive routine with the alarm number. **ALARM** requests a priority display of the error code on the DSKY, overriding any existing display, and also illuminates the PROG[ram Alarm] light. The crew brings up Verb 05, Noun 09 to display the current alarm number, together with those for any other program alarms that might have preceded the restart. Finally, a number of important variables are set back to their initial state. The integrity of the restart phase table is verified, and restart processing for each active phase is performed. Restart groups are handled in a sequence, from group six backwards to group one. After the restart sequences are complete, the previously executing programs are running again and critical data such as IMU orientation and the state vector are preserved. All told, the entire restart process completes in seconds. After acknowledging the alarm by pressing the Reset key, the astronaut can reenter the request, terminate the program, or take other actions.

0.6 Program abort: P00DOO

Program aborts represent more significant problems than program alarms, and result in a P00DOO abort. For example, if a program encounters an unrecoverable logic error such as attempting to take the square root of a negative number, it immediately terminates without invoking the restart processing routines. Erasable storage is not cleared, thus preserving critical system data that is required even with the system idling. With the Major Mode no longer executing, the AGC enters an idling state, displayed as Program (Major Mode) 00. Program 00 reflects the state where no Major Mode is active, yet all other routines (attitude control, telemetry downlink, etc) are actively running. As in the case of program alarms, when the error is detected, the P00DOO Executive routine is called with the abort number. Again, the PROG light is illuminated, and Verb 05 Noun 09 displays the abort code to the astronaut. Program 00 appears as the Major Mode after the Reset key is pressed, and the computer is in a state ready to start a new program.

0.7 Program abort: software restarts

Both a fresh start (hardware restart) and a software restart (often confusingly called simply a “restart”) begin with similar actions to quiesce virtually all system activity. Software restarts focus on bringing previously running program back up, while a fresh start resets the AGC back to an initial idle state, ready to start new work. In both cases, all jobs are canceled and all tasks in the waitlist queue are deleted. Once the system is down, the hardware and software restart routines

continue, but with different objectives. For the recovery process to succeed, the Executive must perform several tasks without distraction, therefore early in the restart sequence all interrupts are inhibited. Timers and flags for the T3RUPT through T6RUPT interrupts are set to prevent their routines from executing until their environment is cleaned up. The three tables associated with units of work the Core Sets, VAC areas, and the waitlist queue - are cleared and reinitialized. DSKY routines are reset to their initial state, and the display is cleared. Warning lights are turned off, with the exception of the PROG and IMU error lights if these are illuminated. Important system variables, possibly containing clues to the nature of the restart, are gathered and downlinked to the ground for analysis. A full dump of erasable storage would be both time consuming and unlikely to provide much additional diagnostic information.

0.8 Rebooting: the AGC fresh start

A fresh start, on the other hand, bypasses the program recovery process and continues shutting the system down to an idle state. Shutdown commands are sent to the SPS, DPS or APS engines if they are firing. Much of the hardware is reset, including the IMU. The loss of the inertial platform requires that the Digital Autopilot be placed into an idle state and all thruster activity terminated. Without an attitude reference, computer control during thruster firings is impossible. Since the AGC is progressing to an idle state, no restarts to recover running programs are run, and the restart tables are reinitialized. Finally, with the hardware and software fully reset, Major Mode 00 (Program 00) is displayed on the DSKY.